CHUNKING

The Decision That Everything Else Inherits From

Building an Agentic RAG System

Part 1

The Problem

"Most RAG tutorials treat chunking as a footnote."

Your embedding model, retrieval precision, re-ranker, and generator quality are ALL downstream of how you split documents.

Get chunking wrong
and no model swap recovers it.

The Naive Approach

Fixed-Size Chunking

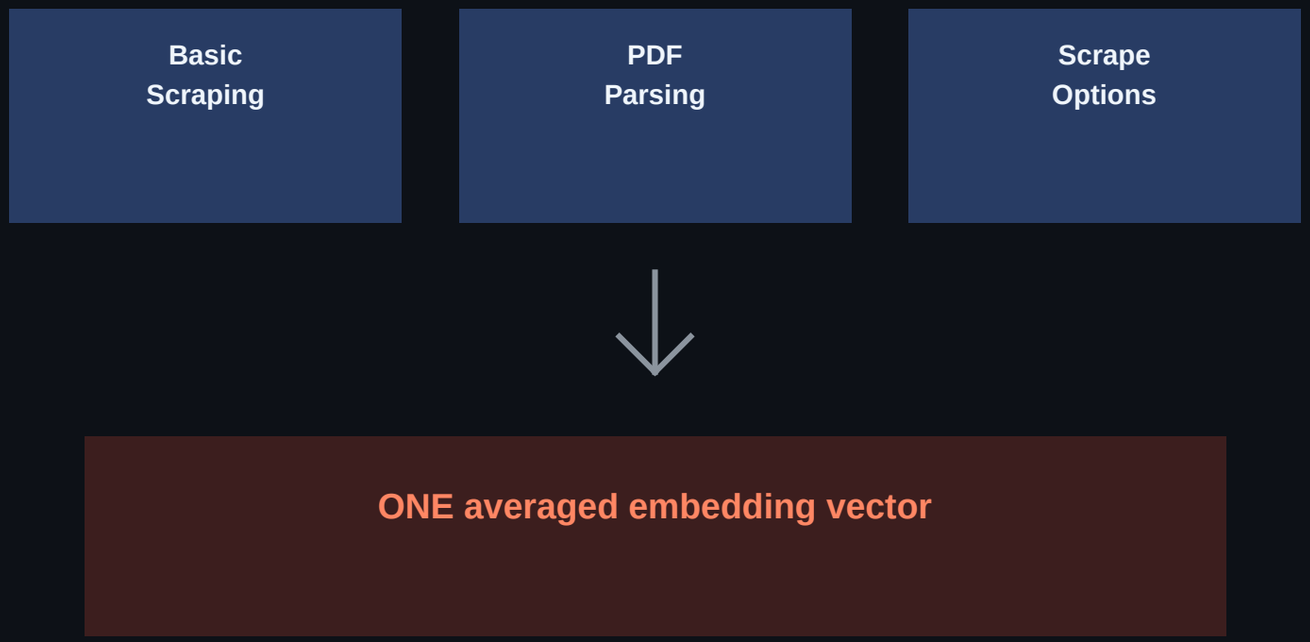
```
chunks = [text[i:i+512]
  for i in range(
    0, len(text), 512
  )]
```

What happens:

- Table split mid-row -- the next chunk has no heading or context
- 3 topics in 1 chunk -- the embedding vector is an average of all three

Topic Dilution

Why averaging kills retrieval



"Ask about PDF parsing specifically
and this chunk scores mediocre --
the answer is averaged out."

Header-Driven Chunking

"A markdown heading is a semantic boundary declared by the author. More reliable than any token heuristic."

Each heading level = a topic boundary:

```
#  = Page topic

## = Section topic

### = Sub-section topic
```

One section = one chunk = one topic
= one clean embedding vector

The Breadcrumb Prefix

```
Section: Advanced Scraping Guide
  > Scrape options
  > Formats
```

Prepended to every chunk before embedding and indexing.

- Richer embedding vectors: model sees the full topic path, not just body text
- Better BM25 tokens: high-signal terms like "scraping", "options", "formats" appear in the document representation

Smart Merging

Handling pure-header chunks

BEFORE

```
Chunk A: ["## Scrape options"]
  (empty -- just a header!)

Chunk B: ["### Formats..."]
  (the actual content)
```

AFTER

```
Chunk B: [
  "## Scrape options",
  "### Formats...",
  "The formats array..."
]
```

No wasted index slots.
No content-free embeddings.

The Results

Metric	Before	After
HR@1	~33%	**66.67%**
HR@5	~58%	**91.67%**
MRR@5	~0.38	**0.78**

Same embedding model.
Same search algorithm.
Only the chunk boundaries changed.

What's Next

Three known failure modes to solve

Level 3 **Giant Sections**

Token overflow -- embedding truncates the tail silently.

Fix: overlapping sliding window within the section.

Level 4 **Cross-Section Answers**

Answer spans two sections -- neither chunk retrieves well.

Fix: inter-section overlap + parent expansion.

Level 5 **No Headers / Code**

Flat files produce one giant chunk. Code blocks embed poorly.

Fix: paragraph fallback + language-tagged code chunks.

Chunking quality sets the ceiling for everything downstream.

"You cannot retrieve what wasn't cleanly embedded."

Follow for Part 2:

BPE vs Whitespace Tokenization for BM25